

**2** Which of the following statements is incorrect?

- A** Measuring the diameter of a wire repeatedly and calculating the average will reduce the absolute uncertainty of the diameter.
- B** Plotting a series of load and extension readings for a spring system on a graph and using gradient to determine spring constant will reduce random error.
- C** Adjusting an ammeter to remove its zero error before measuring a current will remove the systematic error.
- D** Timing a large number of oscillations to find a period will reduce the absolute uncertainty of period.

**3** A boy kicks a ball with an initial velocity of  $12.0 \text{ m s}^{-1}$  at an angle of  $25.0^\circ$  to the vertical